

Dharani Barathi C D

Chennai, Tamil Nadu, India | dharanibarathicd@gmail.com | 8825927230

linkedin.com/in/dharanibarathicd | github.com/iamboffin

Education

Vellore Institute of Technology, Chennai

2023 – 2027

B.Tech in Electrical and Electronics Engineering

Professional Summary

Power electronics enthusiast with hands-on experience in high-gain DC–DC converter design, magnetic component development, PCB layout, and medium-power (200 W, 400 V) hardware validation. Proficient in simulation-driven design (MATLAB/Simulink, PSIM) and lab testing with differential and current probes; experienced in end-to-end converter development from topology selection through hardware debugging.

Patents & Publications

High-Gain DC-DC Power Converter with Cascaded Energy-Transfer Path | *Indian*

Mar 2026

Patent Application

Balaji S, Chada Sai Sravan, **Dharani Barathi C D**, M. Prabhakar App. No. **202641032862** · Pub. No. IN202641032862 A1 · IPO, India

Smart Wheelchair Navigation: Gesture Control and Obstacle Avoidance | *IEEE*

Jan 2025

ICICNIS 2024

Jayasuriya S, **Dharani Barathi C D**, Balaji S, Sriramalakshmi P 10.1109/ICICNIS64247.2024.10823349

Experience

Avionics Engineer, Team Ignition – VIT Chennai (Student Rocketry Team)

Oct 2024 – Apr 2026

- Served as avionics engineer in a multidisciplinary rocketry team; coordinated with structures and propulsion sub-teams on payload envelope, mass budget, and power requirements.
- Designed the full payload flight computer (schematic, PCB, flight integration); collaborated with software members on firmware integration for sensor acquisition and flight data logging.

Intern, Nuclear Desalination Demonstration Plant – BARC, Kalpakkam

May 2025 – Jun 2025

- Analyzed substation SLDs for industrial motor load distribution; studied cable sizing and DC motor protection schemes.
- Designed a VFD schematic for single-phase motor application in EasyEDA; studied SPWM modulation and inverter switching concepts.

Technical Skills

Power Electronics DC–DC Converters (Boost, High-Gain), Open-loop PWM Control, Inductor & Magnetic Design (ETD core), 50 kHz Switching, High-voltage Layout, Thermal Debugging

Simulation & Tools MATLAB/Simulink, PSIM, Altium Designer, EasyEDA

Hardware Validation MDO, Differential & Current Probes, DSO/CRO, Multimeter

Microcontrollers STM32, ESP32, Raspberry Pi Pico (PWM, ADC, UART, SPI, I2C)

Technical Projects

Cubic Gain Boost Converter | 24 V → 400 V | 200 W | 50 kHz | *Design, Simulation,*

Magnetics & HW Validation

- Designed non-isolated high-gain topology using two synchronized switches; modeled in MATLAB/Simulink, cross-verified in PSIM – achieved 93% efficiency at rated load.
- Fabricated 2-layer PCB (25 cm × 15 cm) in Altium Designer with ground plane optimization and power loop minimization.
- Performed full magnetic design and hand-wound three ETD59 core inductors (inductance, ripple current, wire gauge calculations).
- Validated hardware using MDO with differential and current probes; diagnosed thermal stress on low-side MOSFET; topology forms basis of published Indian patent (App. No. 202641032862).

Rocket Electrical Power System | 2S Li-ion | Avionics Payload | *Team Ignition, VITC*

- Designed compact EPS integrating flight computer on a single 8.5 cm × 8.5 cm PCB; implemented regulated 5 V (TPS54528) and 3.3 V (AMS1117) rails from a 2S Li-ion source.
- Incorporated Remove-Before-Flight safety mechanism; designed for vibration resilience and tight form-factor integration.